

Multiple Choice (10 marks)

1. A meteorologist reports that there is a 30% probability of rain and no sun. If there is a 40% probability of no rain, then the probability of both rain and sun is
 - (a) 0.16
 - (b) 0.24
 - (c) 0.3
 - (d) 0.5
2. When $\sqrt[3]{-128}$ is simplified, the result is $a\sqrt[3]{b}$, where a is an integer, and b is a positive integer. If b is as small as possible, then what is $a + b$?
 - (a) 2
 - (b) -4
 - (c) 6
 - (d) -2
3. Simplify 5^3 . Use a calculator, paper and pencil, or mental math.
 - (a) 15
 - (b) 125
 - (c) 8
 - (d) 53
4. Find the quotient of $5.04/0.7$
 - (a) 0.72
 - (b) 0.702
 - (c) 10.0571
 - (d) 7.2
5. Marguerite earned a score between 75 and 89 on all of her previous spelling tests. She earned a score of 100 on her next test. Which of the following statements is true?
 - (a) The mode will increase.
 - (b) The mean will increase.
 - (c) The mean will decrease.
 - (d) The median will decrease.

True / False (10 marks)

Answer True or False

6. True or False: The probability of picking a blue chip from a bag with 10 red chips and 5 blue chips is 2 out of 15.
7. True or False: There are 999 three-digit positive integers, starting from 100 to 998.
8. True or False: The number of rainy days in a given week cannot be modeled by a binomial distribution.
9. True or False: 0.32 grams is equal to 32 centigrams.
10. True or False: The ratio 24 over 64 forms a proportion with the ratio 3 over 8.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Rob feeds his cats using 1 box of cat food every 5 days. Calculate how many boxes he would need for a month, and explain your reasoning and calculations in detail.
12. Discuss the scenario of randomly selecting three points on a circle. Analyze the conditions under which all pairwise distances between these points are less than the circle's radius. What is the probability of this occurring?

End of Paper